

KIM Centroid Generator - User Guide

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This guide will help you use the KIM Centroid Generator application to automatically process radiation therapy DICOM files and calculate seed/ marker positions. This application runs on your computer and monitors a folder for new DICOM files.

Clinical Warning:

DO NOT use this software for clinical purposes unless comprehensive geometric testing has been performed and validated by a qualified medical physicist.

Verification testing must be repeated following any relevant system changes, including but not limited to: treatment machine, treatment planning system (TPS), KIM software, or any other components in the clinical workflow.

Failure to perform appropriate testing may result in inaccurate results with catastrophic clinical risks.

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First-Time Setup

Step 1: Create the Monitoring Folder

Before running the application for the first time, you need to create a folder where DICOM files will be placed.

1. Open **File Explorer** (the folder icon on your taskbar)
2. Navigate to **This PC → Local Disk (C:)**
3. Right-click in the empty space and select **New → Folder**
4. Name the folder exactly: `kim` (lowercase)
5. Your folder path should be: `C:\kim`

Important: The folder name must be exactly `kim` in lowercase.

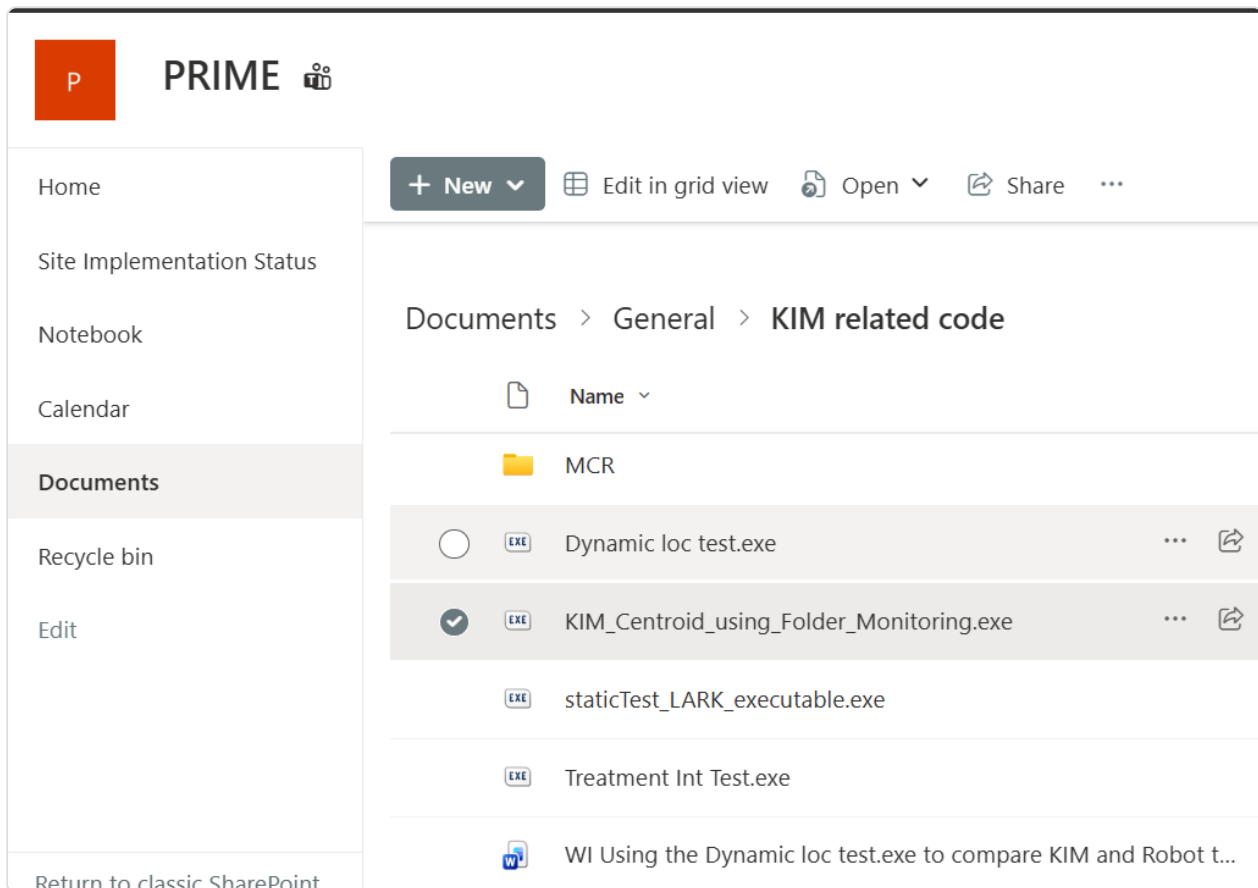
Step 2: Locate the Application

The application file is named:

```
KIM_Centroid_using_Folder_Monitoring VX.X.exe
```

Alert: *Ensure that you have the correct version.*

The correct version of this file is in the [PRIME SharePoint folder](#). If uncertain contact the Coordinating Physicist.



Location of the KIM Centroid Generator executable file in the PRIME SharePoint folder

Step 3: Handle Antivirus Warnings (If Needed)

When you first run the application, Windows or your antivirus software may show a warning. This is normal for new applications.

If Windows Defender shows a warning: 1. Click "**More info**" 2. Click "**Run anyway**"

If your antivirus blocks the file: - Contact your IT department to add this file to the allowed/trusted list - The file is safe - it's a medical imaging processing tool necessary for this trial.

Starting the Application

Step 1: Launch the Application

1. **Double-click** the `KIM_Centroid_using_Folder_Monitoring.exe` file
2. A black window (command prompt) will appear - this is normal!
3. The window shows the application status and activity

Step 2: Choose Interactive Mode

When the application starts, you'll see this prompt:

```
Enable interactive mode for custom structure selection? (y/n):
```

What does this mean?

- **Type `n` and press Enter** - Normal mode (recommended for most users)
 - The application will automatically look for standard structures: seed1, seed2, seed3, au1, au2, au3
 - Best for routine processing
- **Type `y` and press Enter** - Interactive mode
 - If standard structures aren't found, you can manually select which structures to process
 - Best if your files use different naming conventions

For most users, type `n` and press Enter.

Step 3: Verify Monitoring is Active

After selecting your mode, you'll see:

```
Monitoring folder: C:\kim  
Press Ctrl+C to stop monitoring
```

The application is now running and watching the `C:\kim` folder for new DICOM files!

Alert: Keep this window open - the application needs to stay running to monitor for files.

Daily Usage - Processing Files

Once the application is running, follow these simple steps to process DICOM files:

Step 1: Prepare Your DICOM Files

For each patient, you need **two DICOM files**: 1. **RTSTRUCT file** (Radiation Therapy Structure Set) 2. **RTPLAN file** (Radiation Therapy Plan)

Depending on export process these files typically have names like: - `RS.1.2.246.352.71.xxxx.dcm` (RTSTRUCT) - `RP.1.2.246.352.71.xxxx.dcm` (RTPLAN)

Step 2: Copy Files to the Monitoring Folder

1. Open File Explorer and navigate to `C:\kim`
2. **Copy** (or drag and drop) both DICOM files into this folder

The application will automatically detect them.

Step 3: Watch the Processing

In the application window, you'll see messages like:

New DICOM file detected: RS.1.2.246.352.71.xxxx.dcm
Identified as RTSTRUCT for patient: 12345
New DICOM file detected: RP.1.2.246.352.71.xxxx.dcm
Identified as RTPLAN for patient: 12345
Processing complete pair for patient: 12345

```
Default
Structure 'au 2' is not listed in the available ROIs from StructureSetROISequence.
Attempting to find structure: seed3
Structure 'seed3' found in DICOM list of ROIs. Attempting to retrieve contours.
Getting contours for: seed3
Found seed3 with ROI Number: 15
Successfully extracted contours for seed3 (ROI Number: 15)
Successfully retrieved contour points for seed3. Calculating centroid.
Calculated centroid for seed3: [ 13.87962963 -12.64777778 -13.1037037 ]
Attempting to find structure: seed 3
Structure 'seed 3' is not listed in the available ROIs from StructureSetROISequence.
Attempting to find structure: au3
Structure 'au3' is not listed in the available ROIs from StructureSetROISequence.
Attempting to find structure: au 3
Structure 'au 3' is not listed in the available ROIs from StructureSetROISequence.
Finished structure processing loop.
Attempting to retrieve isocenter.
Retrieving isocenter from RT Plan.
Successfully retrieved IsocenterPosition from 3.1, ControlPointIndex 0: [-1.3 0. -4. ]
Isocenter retrieved: [-1.3 0. -4. ]
Constructing output folder: 210119_BeamID_3.1_3.2
Successfully created output directory: C:\kim\210119_BeamID_3.1_3.2
Saving output to file: C:\kim\210119_BeamID_3.1_3.2\Centroid_210119_BeamID_3.1_3.2.txt
Successfully wrote output file: C:\kim\210119_BeamID_3.1_3.2\Centroid_210119_BeamID_3.1_3.2.txt
Moved C:\kim\210119_StrctrSets.dcm to C:\kim\backup\210119_StrctrSets.dcm
Moved C:\kim\210119_KIMFinal.dcm to C:\kim\backup\210119_KIMFinal.dcm
Moved DICOM files to C:\kim\backup
Finished processing pair: RTSTRUCT 'C:\kim\210119_StrctrSets.dcm', RTPLAN 'C:\kim\210119_KIMFinal.dcm'.
Resetting detected files for pair: RTSTRUCT 'C:\kim\210119_StrctrSets.dcm', RTPLAN 'C:\kim\210119_KIMFinal.dcm'. Ready f
or new detection cycle.
```

Console window displaying processing messages.

Step 4: Find Your Results

After processing (usually takes a few seconds), a new folder is created inside
C:\kim :

Folder name format:

PatientID_BeamID_Beam1Name_Beam2Name

Example:

12345_Beam1_BEAM1_BEAM2

Inside this folder, you'll find a text file named:

```
centroids.txt
```

This file contains the calculated positions of the seeds/markers.

Step 5: Processed Files Moved to Backup

After successful processing, the original DICOM files are automatically moved to:

```
C:\kim\backup\
```

Using Interactive Mode (Advanced)

If you enabled interactive mode and the application cannot find standard structures (seed1-3, au1-3), you'll be prompted to select structures manually.

When You'll See This

After placing DICOM files, if standard structures aren't found, you'll see:

```
No default structures found. Available structures in RTSTRUCT:
```

1. PTV
2. CTV
3. GTV
4. Bladder
5. Rectum
6. ProstateSeed_1
7. ProstateSeed_2
8. ProstateSeed_3

```
Enter structure numbers to process (comma-separated, 'all', or 'skip'):
```

How to Select Structures

Option 1: Select specific structures - Type the numbers you want, separated by commas - Example: `6,7,8` (to select ProstateSeed_1, ProstateSeed_2, ProstateSeed_3) - Press **Enter**

Option 2: Select all structures - Type: `all` - Press **Enter** - The application will process every structure in the file

Option 3: Skip this file - Type: `skip` - Press **Enter** - The file pair will be moved to backup without processing

[Screenshot: Interactive structure selection prompt]

Understanding the Results

When you manually select structures, they are labeled as: - Seed 1 (first structure you selected) - Seed 2 (second structure you selected) - Seed 3 (third structure you selected) - And so on...

The results still appear in the `centroids.txt` file in the patient-specific folder.

Stopping the Application

When you're done for the day or need to stop monitoring:

Step 1: Stop Monitoring

1. Click on the application window (the black command prompt window)
2. Press **Ctrl+C** on your keyboard (hold Ctrl, then press C)

You'll see a message like:

```
Monitoring stopped by user
```


Step 2: Close the Window

Simply click the **X** button on the window to close it.

Note: You can also just close the window directly - both methods are safe.

Understanding Your Results

The Centroids File

The `centroids.txt` file contains position information in this format:

```
Patient ID
LastName,FirstName
Seed1, X= x.xx1, Y= y.yy1, Z= z.zz1
Seed2, X= x.xx2, Y= y.yy2, Z= z.zz2
Seed3, X= x.xx3, Y= y.yy3, Z= z.zz3
Isocenter (cm), X= x.xx, Y= y.yy, Z= z.zz
```

For example:

```
210119
zzKIMe2e,zzKIMe2e
Seed1, X= -1.63, Y= -1.18, Z= 0.69
Seed2, X= -0.09, Y= 1.25, Z= -0.28
Seed3, X= 1.39, Y= -1.26, Z= -1.31
Isocenter (cm), X= -0.13, Y= 0.00, Z= -0.40
```

What to Do With This File

- KIM software will require this file to be provided before tracking
 - Prior to using at your centre ensure that appropriate testing has been conducted to ensure geometric accuracy and compatibility.
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Troubleshooting

Problem: Application Won't Start

Symptom: Nothing happens when you double-click the .exe file

Solutions: 1. **Check antivirus:** Your antivirus may have blocked or quarantined the file - Open your antivirus software - Look for quarantined files - Restore the file and add it to the allowed list - Contact IT if you need help with this

1. **Check file location:** Make sure you're clicking the correct .exe file
2. The file should end in `.exe`
3. File size is approximately 17 MB

Problem: "Permission Denied" or "Access Denied" Error

Symptom: Error messages about file or folder permissions

Solutions: 1. **Check folder permissions:** - Right-click on `C:\kim` folder - Select **Properties** → **Security** tab - Ensure your user account has "Full control" - Contact IT if you need permissions changed

1. **Run as Administrator (temporary solution):**
2. Right-click the .exe file
3. Select **"Run as administrator"**
4. This should only be needed if there are permission issues

Problem: Files Not Being Processed

Symptom: DICOM files placed in C:\kim but nothing happens

Solutions: 1. **Check if application is running:** - Look for the black command prompt window - If it's closed, restart the application

1. **Verify both files are present:**
2. You need BOTH an RTSTRUCT and RTPLAN file for the same patient
3. Check that Patient IDs match in both files

4. **Check file readiness:**

5. If copying large files over a network, wait 30 seconds after copying
6. The application waits for files to finish transferring

7. **Look for error messages:**

8. Check the black window for red error messages
9. Take a screenshot and send to IT support

Problem: Can't Find Results

Symptom: Processing completed but can't find the output folder

Solutions: 1. **Check inside C:\kim:** - Open File Explorer - Navigate to `C:\kim` - Look for folders with patient ID in the name

1. **Check the application window:**

2. Look for messages showing where results were saved
3. The path will be shown after "Created patient output folder:"

Problem: Application Window Closes Immediately

Symptom: Black window appears and disappears instantly

Solutions: 1. **Missing monitoring folder:** - Make sure `C:\kim` folder exists
- Create it if it doesn't exist (see First-Time Setup)

1. **File is corrupted:**

2. Try downloading the .exe file again
 3. Download fresh copy from PRIME directory and try again.
 4. Try it on a different laptop/workstation.
 5. Get someone else to try it.
 6. Call trial physicist and ask for another file to be regenerated.
-

Getting Help

Who to Contact

Contact the KIM WG or authors of this software (KRM, KK) with: - Software version. - Description of the problem - Screenshots of error messages - The DICOM files you were trying to process.

Information to Provide

When reporting an issue, include: - Your name and department - What you were trying to do - What happened instead - Any error messages (with screenshots) - Patient ID (if relevant and appropriate to share)

Quick Reference Card

Starting the application: 1. Double-click

KIM_Centroid_using_Folder_Monitoring.exe 2. Choose interactive mode (type `n` for normal, `y` for advanced) 3. Keep window open

Processing files: 1. Copy RTSTRUCT and RTPLAN files to `C:\kim` 2. Wait for processing (watch the application window) 3. Find results in new patient folder inside `C:\kim`

Stopping the application: 1. Press Ctrl+C in the application window, or 2. Close the window

Finding results: - Location: `C:\kim\PatientID_BeamID_Beam1_Beam2\` - File name: `centroids.txt`

Getting help: - Contact KIM Working Group, Authors

This guide was created for clinical users of the KIM Centroid Generator application. For technical documentation, developers should refer to README.md. in the repository

